TAAPU

TEST FOR ASSESSMENT OF ARTICULATION AND PHONOLOGY IN URDU

By

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INTRODUCTION

Test for articulation and phonology in Urdu (TAAPU) is a single word test that was designed for assessment of speech sound deficits in preschool and school age children who visit speech therapy clinic with articulatory and phonological disorders speaking Urdu language in Pakistan. It assesses all consonantal sounds at different positions in words of Urdu language. It is used by speech therapists working in clinical and educational settings to find out sound errors of substitution, omission and addition and determines the phonological disorder. Time to administer TAAPU is approximately 30 minutes. This supplemental manual includes an overview of the phonology section, administration and scoring procedures, and normative data for Phonological Checklists I and II. Attached in the manual are background information form, oral motor examination guide, word response checklist, phonological analysis checklist and Urdu phonetic alphabets.

TAAPU Application:

TAAPU is available in soft and hard form both. It is applied on individuals with speech sound disorders between ages 4-8 years. To differentiate between articulatory or phonological disorders a step wise method has been adopted for the scoring of responses.

Step: 1

Back ground information will be sought to determine the probable cause and determination of compliance and attitude towards therapy. Background information form is provided in the manual.

Step: 2

An oral motor examination will be conducted using oral motor examination guide to rule out any pathology that may be causing speech sound disorder.

Step: 3

While rapport will be built with the child using general conversation, singing poem, talking about some recent or fun activity or counting child's speech will be informally assessed for intelligibility. A story

picture will be shown to get sample of continuous speech to find out speech sound error at conversational level.

Step: 4

Child will be seated in a manner that his attention does not divert. After preparing the child to cooperate during test, he will be guided to name the pictures on flash cards or on lap top. Child will name the pictures one by one spontaneously and if he will not be able to name spontaneously, ST will make the child imitate production. Spontaneous production is first priority imitation can be used in case word seems unfamiliar to the child or code switching occurs. Responses of the child will be recorded on response sheet simultaneously or could be audio/video recorded to analyze sample later and fill the response sheet using phonetic transcription.

Step: 5

Responses will be noted in the form of whole word utterance in the form of misarticulated words or code switched words. In case of code switching child will be asked for Urdu noun if he is unable to recall then imitative production will be attempted. Child's responses will either be audio or video recorded or child's production can be written on response sheet for analysis. After analysis errors will be noted on the sheet in the form of substitution, omission, addition of single sound at initial, middle and end position in the words/responses.

As one sound error is detected multiple times due to occurrence of same sound in various words, the phonological process can easily be detected. Internationally phonological analysis is carried out by deriving the percentage calculating frequency of occurrence of same speech sound error. Hence phonological error will be determined as shown in the tables of phonological analysis. The 60% occurrence of errors in one phonological pattern will confirm the disorder.

PSYCHOMETRIC CHARACTERISTICS OF TAAPU

The technical characteristics of TAAPU were determined based on the nine criteria suggested by Mc Cauley and Swisher (1984) and 11 criteria recommended in a graduate study at Ohio State University (2010) "psychometric review of speech test for preschoolers: 25 years later". the table below present those characteristics:

Criterion No.	Criteria	Statistical measures	Conclusion/Remarks
1.	**Test manual with psychometric properties**	**Available in kit**	**--**
2.	**Sample Size**	**140 children with APD**	**35 from each province**
3.	**Chronbach's Alpha reliability**	**.899**	**Whole sample**
4.	**Evidence of concurrent validity**	**NA**	**NA**
5.	**Measure of central tendency and variability**	**Validity and reliability analysis**	**--**
6.	**Evidence of predictive validity**	**Gave same error type**	**--**
7.	**Estimate of test retest reliability**	**.991 paired samples correlation**	**Conducted during development phase**
8.	**Evidence of inter-examiner reliability**	**Cohen's Kappa value .61-.80**	**Substantial agreement among raters**
9.	**Detailed administration procedure**	**Manual**	**Available in kit**
10.	**Clinician qualification**	**PGDSLT, MS or M.Phil**	**--**
11.	**Construct validity**	**Factor analysis**	**--**

Background Information Form

1. **Name ----------**
2. **Age& Gender----------**
3. **Birth order----------**
4. **Study grade----------**
5. **Mother language:----------**
6. **Family structure: Joint family------ nuclear family------single parent family-----others----------**
7. **Socio-economic level:**
   Income level: Mother ----------Father----------
   Educational Level: Mother ----------Father----------
   Occupation: Mother ----------Father----------
8. **Referred by----------**
9. **Contact details----------**

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10. **Case History----------**

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11. **Hearing Status----------**
    ----
12. **Oral Motor Mechanism**

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13. **Speech Intelligibility----------**
14. **Additional comments----------**
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15. **Recommendation**

ORAL MOTOR EXAMINATION FORM

Target	Elicitation	Observations
Facial Symmetry	Have subject look straight at you	Are all facial features symmetrical?
Lips	Smile Pucker Open Close tight Puff up cheeks, hold against resistance	Look for symmetrical movements, extent of movements, ability to close lips completely, and hold against resistance
Tongue	Stick straight out Left Right Left and Right quickly Elevation Retraction Protrusion & lateral movement against resistance (with tongue blade)	Look for symmetry and range of motion. Look at size of tongue and for evidence of fasciculation. Assess ability to exert force against resistance. Assess coordination of fast movements
Oral Cavity	Open wide	Look at teeth, tongue, hard and soft palate. Look for evidence of lesions, atrophy, or missing structures. Also observe jaw excursion, listen for "clicks," watch for lateral shifts
Velum	Say "ah" Elicit gag reflex by touching anterior faucial arches, posterior faucial arches, or base of tongue. Elicit gag on both sides	Watch for extent and symmetry of velar elevation and pharyngeal constriction.
Voice	Maximum Phonation Time s:z ratio Pitch glides	Listen for vocal intensity, pitch, and quality. Note especially breathiness or gurgles

URDU PHONOLOGICAL ANAYSIS SHEETS

FRONTING		
S.NO	**Words**	**Error**
1.	بکری	
2.	کتاب	
3.	پتنگ	
4.	مکھی	
5.	غبارہ	
6.	بطخ	
7.	خربوزہ	
8.	غبارہ	
9.	ناک	
10.	کتاب	
Total score	10	
%age		
Phonological process	Present/absent	

***check all back sounds**

STOPPING		
S.NO	**Words**	**Error**
1.	شیر	
2.	بادشاہ	
3.	بارش	
4.	سورج	
5.	رسی	
6.	گلاس	
7.	خربوزہ	
8.	زبان	
9.	میز	
10.	سانپ	
Total score	10	
%age		
Phonological process	Present/Absent	

***Check all fricatives**

/r/ PHONOLOGICAL DISORDER IN URDU		
S.NO	**Words**	**Error**
1.	رسی	
2.	بارش	
3.	جراب	
4.	بکری	
5.	شیر	
6.	بکری	
7.	مگرمچھ	
8.	مرغی	
9.	مچھر	
10.	چھتری	
Total score	10	
%age		
Phonological process	Present/Absent	

FINAL CONSONANT DELETION		
S.NO	**Words**	**Error**
1.	شیر	
2.	گیٹ	
3.	جراب	
4.	برف	
5.	سانپ	
6.	ویگن	
7.	کتاب	
8.	بارش	
9.	میز	
10.	دانت	
Total score	10	
%age		
Phonological process	Present/Absent	

***check final consonants**

NASAL ASSIMILATION		
S.NO	**Words**	**Error**
1.	میز	
2.	بندر	
3.	ناک	
4.	انگور	
5.	پانی	
6.	چاند	
7.	یونیفارم	
8.	سانپ	
9.	زبان	
10.	پتنگ	
Total score	10	
%age		
Phonological process	Present/Absent	

***check non nasal sounds in words where there is a nasal sound in that word**

DEPALATALIZATION		
S.NO	**Words**	**Error**
1.	ٹماٹر	
2.	گیٹ	
3.	ڈبہ	
4.	جھنڈا	
5.	چڑیا	
6.	پہاڑ	
7.	شیر	
8.	بارش	
9.	بادشاہ	
10.	کارڈ	
Total score	10	
%age		
Phonological process	Present/Absent	

***check palatal sounds**

All speech sounds are mastered by age 8 years, check, if the errors are persistent after the age of mastery, plan therapy or refer child to other professional (ENT, Audiologist, Psychologist, Neurologist) based on the results of TAAPU.